

An example of FAIR data tool: NOMAD, hands-on training



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An example of FAIR data tool: NOMAD, hands-on training material:

Slides

https://github.com/TheOldGogo/FAIRmat-tutorial-16/tree/main/tutorial_16_materials/all_slides



Menti

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Tutorial + example files:

<https://github.com/TheOldGogo/FAIRmat-tutorial-16/blob/main/README.md>

Or

<https://fairmat-nfdi.github.io/FAIRmat-tutorial-16/index.html>



Create an account ----->

<https://nomad-lab.eu/nomad-lab/>



What do we do today?

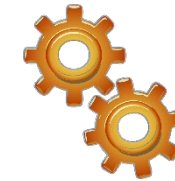


<https://github.com/TheOldGogo/FAIRmat-tutorial-16/blob/main/README.md>

- ❑ Introduction to FAIR data and NOMAD
- ❑ Exploring NOMAD entries, use NOMAD data
- ❑ Upload data and create datasets
- ❑ FAIR Data automation and examples



hands on



hands on



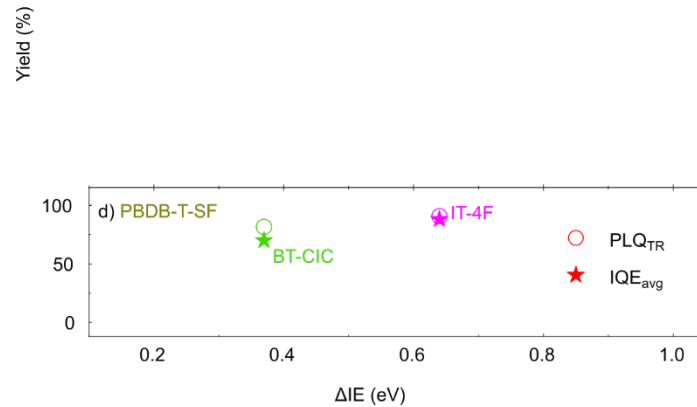


Part 1-

INTRODUCTION TO FAIR DATA AND NOMAD

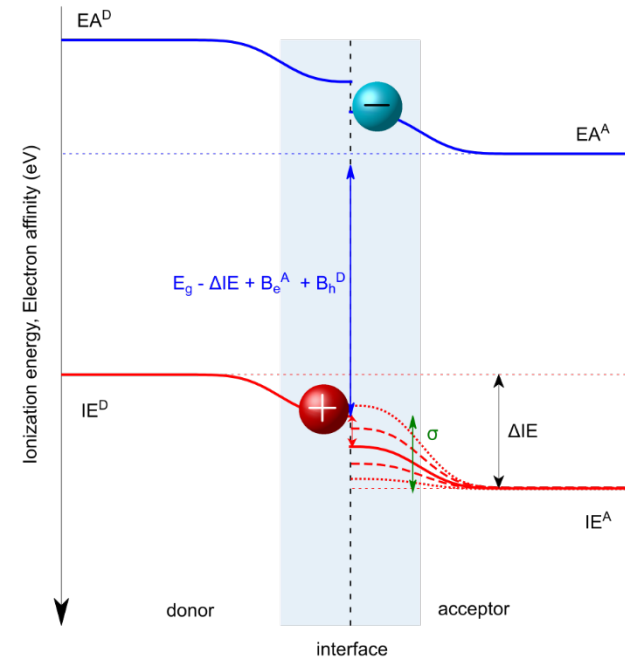
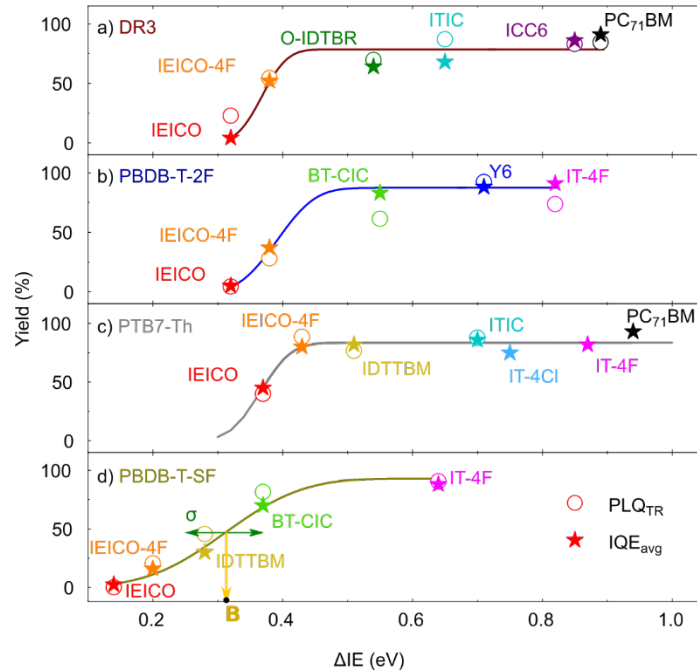
Why FAIR data? Raise impact and enable AI.

Data usable across projects, across laboratories and across platforms.



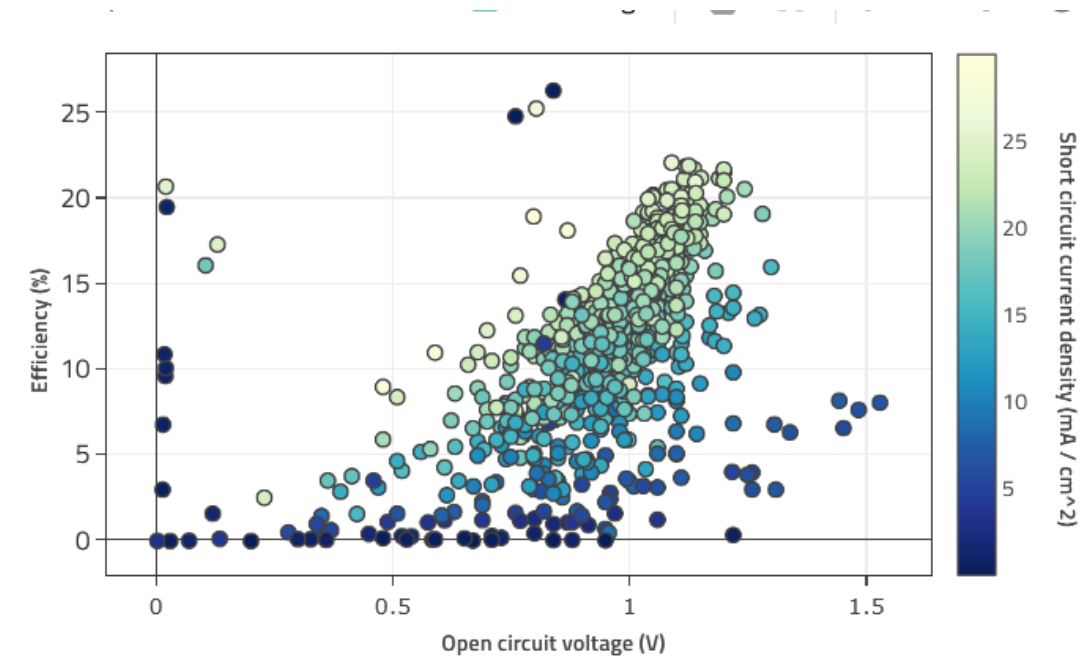
Why FAIR data? Raise impact and enable AI.

Data usable across projects, across laboratories and across platforms.



Why FAIR data? Raise impact and enable AI.

Data usable across projects, across laboratories and across platforms.



Home

3

Search for samples

4

Q

Add sample

Group: [All] Modified: [Any time]

If you are using those tools please don't forget to cite us !
The OHS NMR repository: An integral solution to control the flow of your data from the magnet to the public.
Philip L. Zasso M. Kostro D. Berni A. Castello AM. Solari A. Alencio MA. Pellet N. Toth M. Schöner N. Kuhn S. Magnetic Resonance in Chemistry. 2017.

Double click to open a sample, one click to select it

Reference & Meta

Other information

ethybenzene
final JACS2019
seeTubeID 10452246232
country Colombia
abc
CHIM.1234
admin - created: 2017-06-19 - modified: 2022-07-25
nb1h: 1 nb2d: 4 nbR: 1
nbRaman: 2 nbMass: 3
nbChromatogram: 7
nbUV: 4 nbTGA: 5
nbDSC: 2 nbXPS: 5
nbXPS: 11 nbXray: 6
nbPelletHardness: 1
C₈H₁₀ (106.17)

5

YTC 326
initial 1.5
ISO 1
abc 1
admin - created: 2019-04-04 - modified: 2021-04-28
nbR: 15 nbXPS: 1

PEG
label polymer
label PEG
ipatiny - created: 2020-04-23 - modified: 2020-12-14
nbMass: 1

YTC 0189
admin - created: 2019-04-04 - modified: 2020-11-06
nbR: 15

General

NMR

1

Mass & EA

Cheminformatics

Biology

Tools

Number of samples

Electronic Notebook

Structure search

Open/edit sample

Lipinski

507

IR spectra

IR prediction

Raman spectra

UV spectra

Cyclic voltammetry

IV curves

Differential scanning calorimetry

Thermogravimetric analysis

Differential Sedimentation

X-ray photoelectron spectroscopy (XPS)

Hg porosimetry

Isotherm analysis

Oil absorption number

Pellet hardness

X-ray Fluorescence

DSC

TGA

DCS

XPS

Hg

Isotherm

OAN

PH

XRF

Powder XRD

Xray structure

Image analysis

3D model

Property explorer

Pubchem lookup

Report

PXRD

Xray

IA

3D

Info

Pubchem

Report

2

6

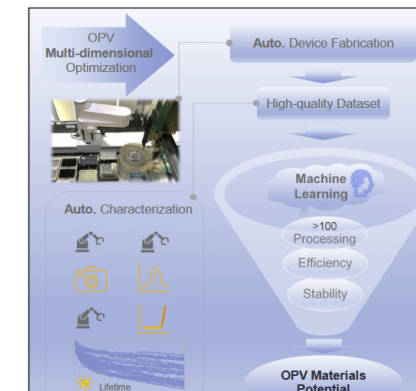
A Bright Future!

Joule

CellPress

Article

Elucidating the Full Potential of OPV Materials Utilizing a High-Throughput Robot-Based Platform and Machine Learning



Xiaoyan Du, Larry Luer, Thomas Heumüller, ..., Tobias Stubhan, Jens Hauch, Christoph J. Brabec
x.du@fu-juellich.de (X.D.)
christoph.brabec@fau.de (C.J.B.)

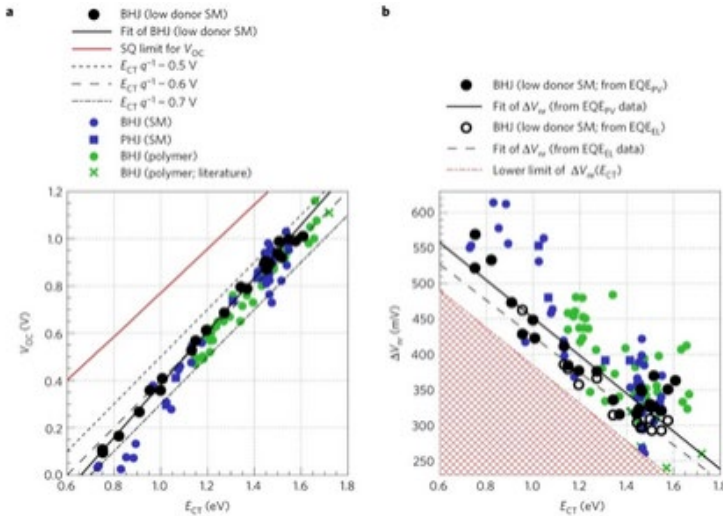
HIGHLIGHTS

Automated robot-based organic solar cell fabrication and characterization

A high-quality dataset enables machine-learning-based predictions

Reliable screening of more than 100 processing variations within 70 h

A platform and methodology for broader research areas in solution-processed devices



nature
energy

ARTICLES

PUBLISHED: 10 APRIL 2017 | VOLUME: 2 | ARTICLE NUMBER: 17053

Intrinsic non-radiative voltage losses in fullerene-based organic solar cells

Johannes Benduhn^{1*}, Kristofer Tvingstedt^{2*}, Fortunato Piersimoni³, Sascha Ullbrich¹, Yeli Fan^{4†}, Manuel Tropiano⁵, Kathryn A. McGarry⁶, Olaf Zeika¹, Moritz K. Riede⁷, Christopher J. Douglas⁶, Stephen Barlow⁴, Seth R. Marder⁴, Dieter Neher³, Donato Spoltore¹ and Koen Vandewal^{1*}

Organic solar cells demonstrate external quantum efficiencies and fill factors approaching those of conventional photovoltaic technologies. However, as compared with the optical gap of the absorber materials, their open-circuit voltage is much lower, largely due to the presence of significant non-radiative recombination. Here, we study a large data set of published and new material combinations and find that non-radiative voltage losses decrease with increasing charge-transfer-state energies. This observation is explained by considering non-radiative charge-transfer-state decay as electron transfer in the Marcus inverted regime, being facilitated by a common skeletal molecular vibrational mode. Our results suggest an intrinsic link between non-radiative voltage losses and electron-vibration coupling, indicating that these losses are unavoidable. Accordingly, the theoretical upper limit for the power conversion efficiency of single-junction organic solar cells would be reduced to about

nature
materials

Check for updates

ip
lls



What FAIR data? Findable, Accessible, Interoperable, Re-usable



Findable

🔍 Data have a place to live + **persistent unique identifiers** + referencing + metadata -> searchable



Accessible

📁 (Open formats), **licensing**



Interoperable

🔗 **Standards**, vocabularies, and APIs



Reusable

🔄 **Documentation**, provenance, and quality





More about FAIR data

More presentation to come on FAIR data in KAUST:

Library training week:

[October 15](#): FAIR data and

Research Data Management:

the future of data sharing



Publication Ethics and Authorship.

[September 30](#):



The tools: Now, hands – on training:

[Slides](#)



**FAIRmat Tutorial 16: Getting started with
NOMAD for materials-science research
data management**





How FAIR data?



(among others)

🔍 Data have a place to live + **persistent unique identifiers** + referencing + metadata -> searchable

🔍 Storage place + searchable + PUI

📁 (Open formats), **licensing**



📁 Proper referencing and **licensing**

🔗 **Standards**, vocabularies, and APIs

🔗 Electronic Laboratory Notebooks
+ standardized info fields

🔄 **Documentation**, provenance, and quality

🔄 APIs for direct calls NOMAD (by
AI or others)





So let's get started! Open NOMAD.



PUBLISH ▾ EXPLORE ▾ ANALYZE ▾ ABOUT ▾

Entries ?



LOGIN / REGISTER



UNITS

Browse existing
data

Use the data

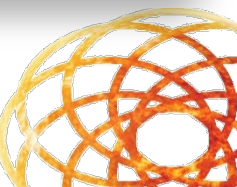
- Visualization
- Statistics
- AI

(not today sorry)

Upload, organize
and publish data



Let's go to [Discord](#),
[click here!](#)





Part 2-

EXPLORE DATA

What we would do in this part

- ❑ Search entries

- ❑ Personalize your dashboard, and visualize results



Resources

- [Step by step guide.](#)



- Original video on [youtube.](#)



Question 1:

How many entries of hexagonal boron nitride (h-BN) are available in NOMAD?

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Hints:

- ☐ Use Elements / Formula filter
- ☐ Use Structure / Symmetry filter



Question 2:

What is the range of bandgap of h-BN in NOMAD entries?

Hint:

☐ Use Properties -> Electronic filter

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Question 3:

How many entries in NOMAD have calculated bandgap of h-BN between 5.9 eV and 6.5 eV?

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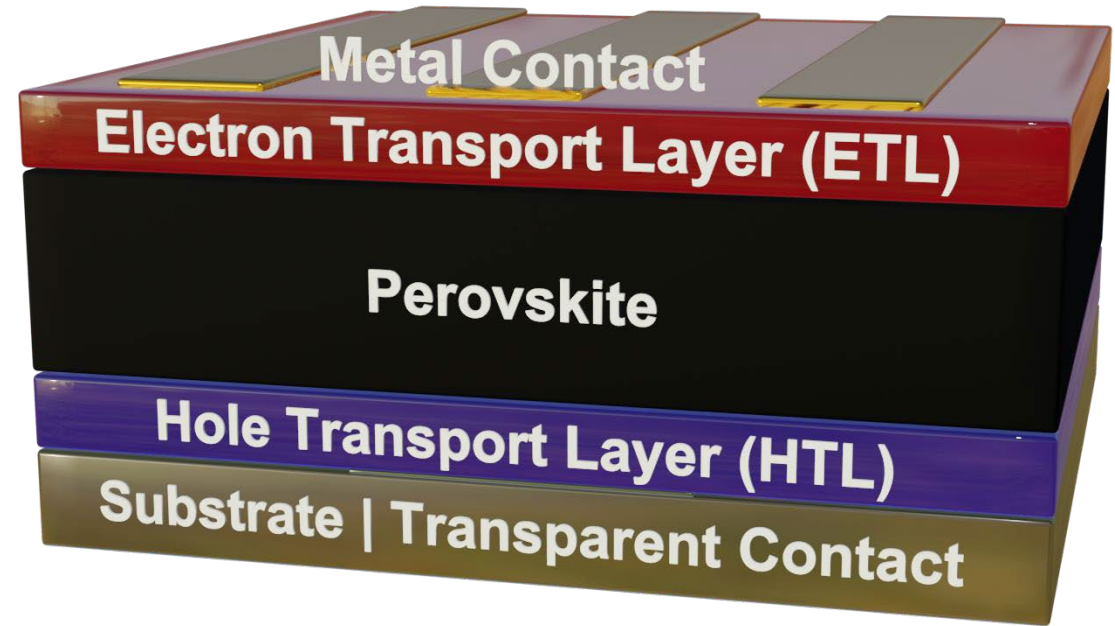
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Hint:

- ☐ Use the slider or give exact min. and max. values in Properties - Electronic filter



Example 2: Solar Cells



Example 2: Solar Cells

My device:

contact: Au

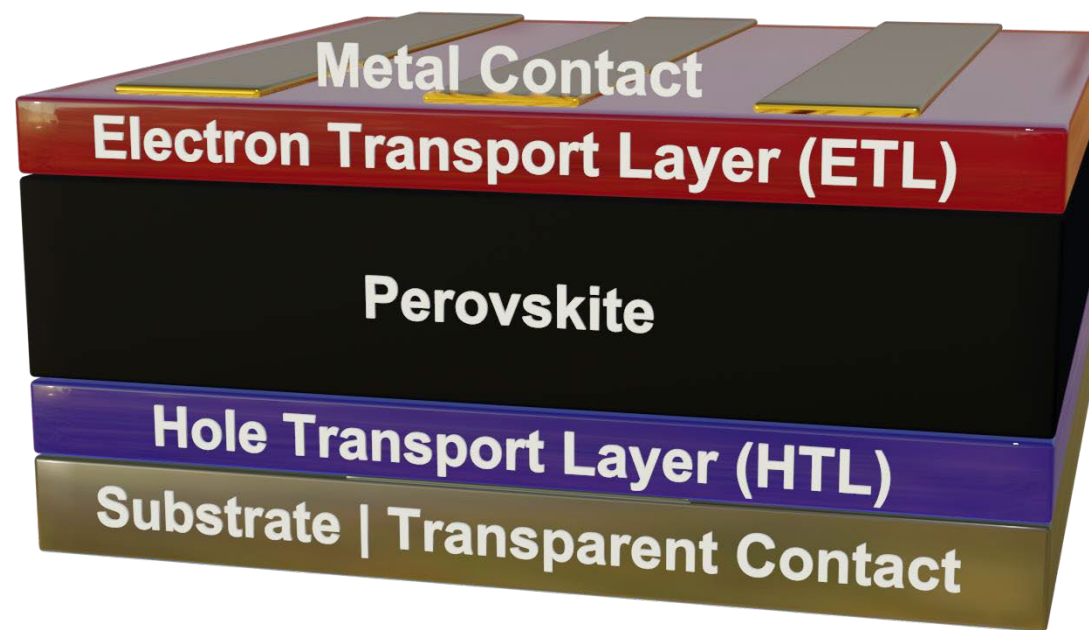
HTL: Spiro-OMeTAD ($C_{81}H_{68}N_4O_8$)

Perovskite absorber: $CsPbBr_2I$

ETL: TiO_2 -c (compact Titanium Dioxide)

contact: FTO (Fluorine-doped Tin Oxide)

Substrate: SLG (Soda Lime Glass)



Question 1:

How many entries in NOMAD have the elements of the absorber layer (CsPbBr₂I)?

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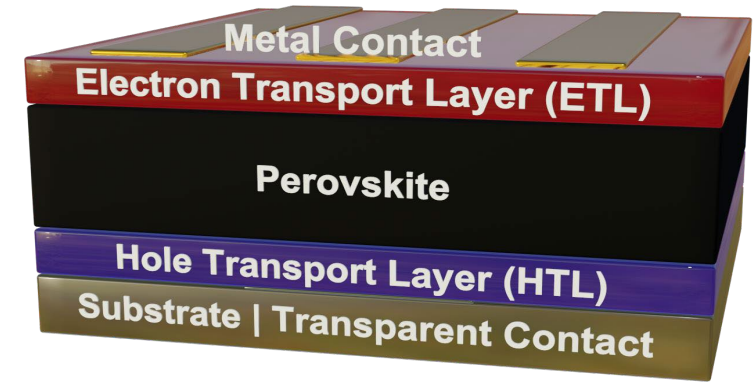
Question 2:

How many of those entries include **solar cells** data with:

- CsPbBr₃ as an **absorber layer**,
- SpiroOMETAD as **Hole Transport Layer** **and**
- TiO₂-c as **Electron Transport Layer** ?

Hint:

☐ Look Use Cases -> Solar Cells filter



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Question 3:

Show the range of reported efficiencies from these entries.

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Hint:

- ❑ Use HISTOGRAM widget, for X AXIS start typing “efficiency”

Question 4:

Show an overview of the available entries with info about **short circuit current density**, **efficiency**, and **open circuit voltage**.

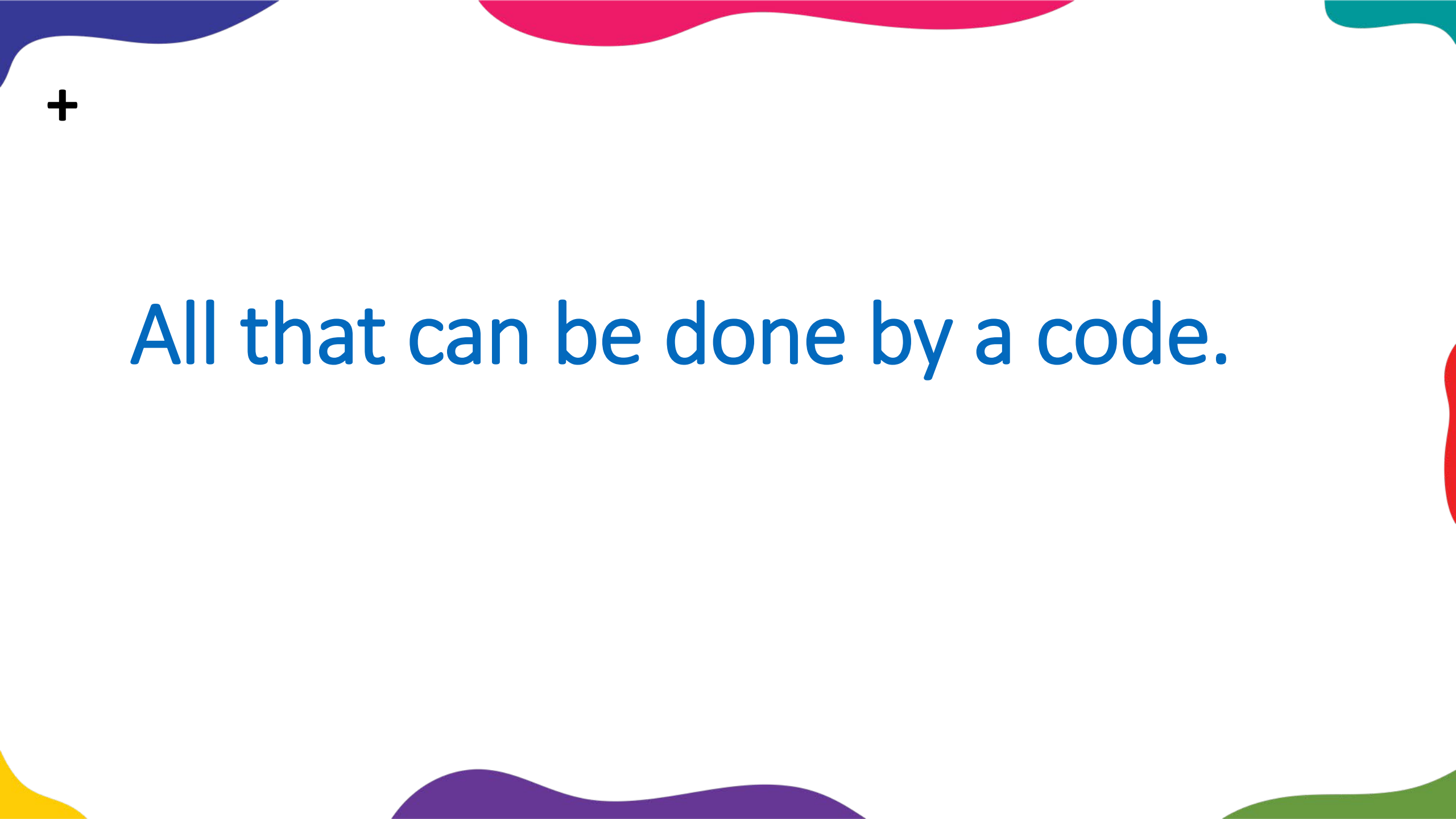
Menti

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Hints:

- ☐ Use **SCATTER PLOT widget**,
 - ☐ for X AXIS start typing “open circuit voltage”,
 - ☐ for Y AXIS -> “efficiency”,
 - ☐ for MARKER COLOR -> “short circuit current”





+

All that can be done by a code.

Summary about exploration

Think of the power of having your data
– together with the rest of the world's –
in something like that.



Sooo.... Part 3-

HOW DO WE PUT OUR DATA IN THERE?

From raw files to published datasets



FAIRmat tutorial 16
February 26, 2025



Publishing your data with NOMAD

To begin the journey from uploading raw files to publishing datasets with a Digital Object Identifier (DOI), you need to understand the **key NOMAD elements involved in the process.**



NOMAD user

To create a user account, follow the steps on this [link](#).



steps to create an account

0- a bit of vocabulary / concepts

- Uploads <- a group of file you upload
- Projects <- your work out of NOMAD
- Entries <- you've already seen that one =
files understood by NOMAD
- Datasets <- A group of related entry, possibly with a DOI

NOMAD uploads

NOMAD user



uploads



project 1



project 2



project 3

- Research data in NOMAD are organized inside uploads.
- Upload can be created for each project and be organized in nested folder directories.
- Each upload can contain multiple files and entries.

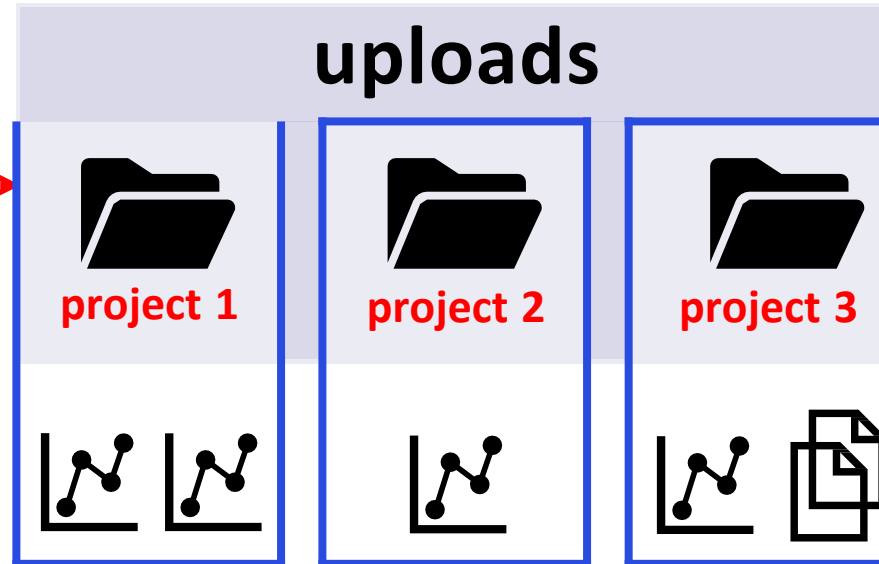
NOMAD entries

NOMAD user



entries

uploads



- Entries are generated automatically by NOMAD.
- Entries can be searched and viewed through dedicated pages.
- Files are processed into entries if a built-in parser is available.

Datasets in NOMAD

NOMAD user



entries

uploads



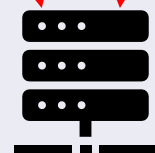
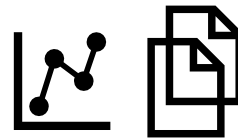
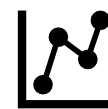
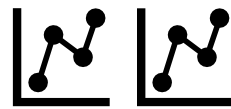
project 1



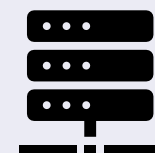
project 2



project 3



dataset 1



dataset 2

datasets

Datasets group related entries either from one upload or several uploads.

The key elements in NOMAD

NOMAD user



entries

uploads



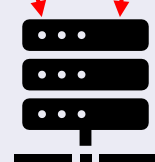
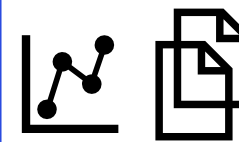
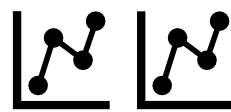
project 1



project 2



project 3



dataset 1



dataset 2

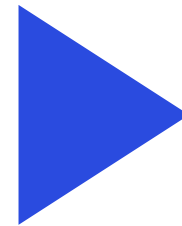
datasets



So let's go: creating and managing uploads

Hands-on tasks:

- ☐ Create your first upload in NOMAD.
- ☐ Rename your upload.
- ☐ Understand the elements and views of the upload page.
- ☐ Share your upload with a collaborator.
- ☐ Publish your upload.



Lets go to NOMAD,
click [here](#)!

Uploading raw files to NOMAD

Hands-on tasks:

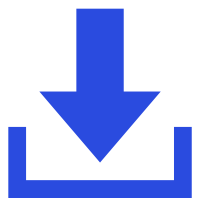
- ☐ Add files to your upload.
- ☐ Distinguish between generated entries and stored files.
- ☐ Preview and visualize your uploaded files/entries.

Exercise 1: uploading miscellaneous files

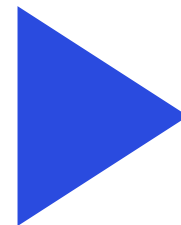
In this exercise, we will use files that are not processed into entries in NOMAD, such as, .pdf, .jpg, .txt, and .csv.

Hands-on tasks:

- ☐ Add files to your upload by drag-and-drop and by file browser.
- ☐ View and organize your files in your upload.
- ☐ Preview the files contents in the GUI.



Download the files!
click [here](#)!



Lets go to NOMAD,
click [here](#)!

Uploaded raw files: entries vs. files

NOMAD Entries

- ☐ Originate from files for which a **built-in parser** in NOMAD exist.
- ☐ NOMAD **extracts the data** based on the data schema.
- ☐ This allows for **searchability, analysis and generating visualization**.

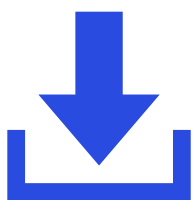
Other Files

- ☐ Originate from files that **do not have a built-in parser** in NOMAD.
- ☐ NOMAD cannot automatically extract and structure their data.
- ☐ They are **stored** as part of your upload, and can be downloaded or shared.

Exercise 2: uploading computations data files

In this exercise, we will use input and output files of a DFT calculation for Iron(III) Oxide.

- These calculations were preformed using a code that is supported by NOMAD, i.e. the FHI-aims code.
- NOMAD has a parser for the FHI-aims code. This means it will create an entry for these data.



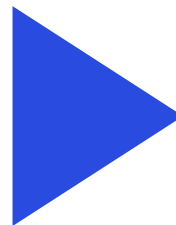
Download the files!

click [here](#)!

Exercise 2: uploading computations data files

Hands-on tasks:

- ☐ Add files to your upload by drag-and-drop.
- ☐ Explore the generated entry page from the files.
- ☐ Provide additional metadata to the entry.

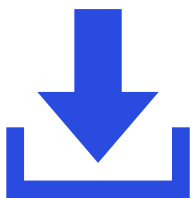


Lets go to NOMAD,
click [here](#)!

Exercise 3: uploading experimental data files

In this exercise, we will upload files of an xray photoelectron spectroscopy (XPS) measurement.

- NOMAD supports .nxs experimental data files.
- Scientific instruments generate raw files in other formats, such as .xml (from SPECS spectrometer).
- NexusDataConverter converts common formats to the .nxs format.



Download the files!

click [here](#)!

Exercise 3: uploading experimental data files

In this exercise, we will upload files of an xray photoelectron spectroscopy (XPS) measurement.

Hands-on tasks:

- ☐ Add .nxs files to your upload by drag-and-drop.
- ☐ Explore the generated entry page from the files.
- ☐ Use the NexusDataConverter to upload files with other formats, such as .xml.



Lets go to NOMAD,
click [here](#)!

Creating and publishing datasets

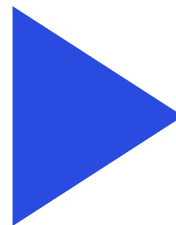
- Datasets are used to group entries, making it easier to manage related data.
- Datasets can include entries from several uploads.
- The same entry can be in several datasets.
- Datasets have no influence on the processing of data.
- Users can get a DOI for their datasets.



Creating and publishing datasets

Hands-on tasks:

- ☐ Create a new dataset, and include entries into the dataset.
- ☐ Explore and manage datasets.
- ☐ Assign a DOI to your datasets.



Lets go to NOMAD,
click [here](#)!



BREATHE



Part 4 -

FAIR DATA AUTOMATION AND EXAMPLES

So... what do you think?

Cool... but...

Overwhelmed?

Laborious?

Complicated?

So long for only one piece of data?

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Yes.... Because this is just the introduction

What's next?

Electronic Laboratory Notebooks

Parsers

GUI

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Guiding through the steps for documenting research :
using NOMAD ELN functionality - Siamak Nakhaie

Parser

Just a line of code

Result:

Setup description:

Credit: Maha Alamoudi

Experimental details:

Sample:

User:

Blabla.vms

Blabla.yaml

NeXus

Look-up Dictionary

Application definition

Parser

NeXus file

Hierarchical structure, F.A.I.R. compliant

Facility

Raw data

Raw metadata

Parser

Just a line of code

Result:

Setup description:

Credit: Maha Alamoudi

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Look-up Dictionary

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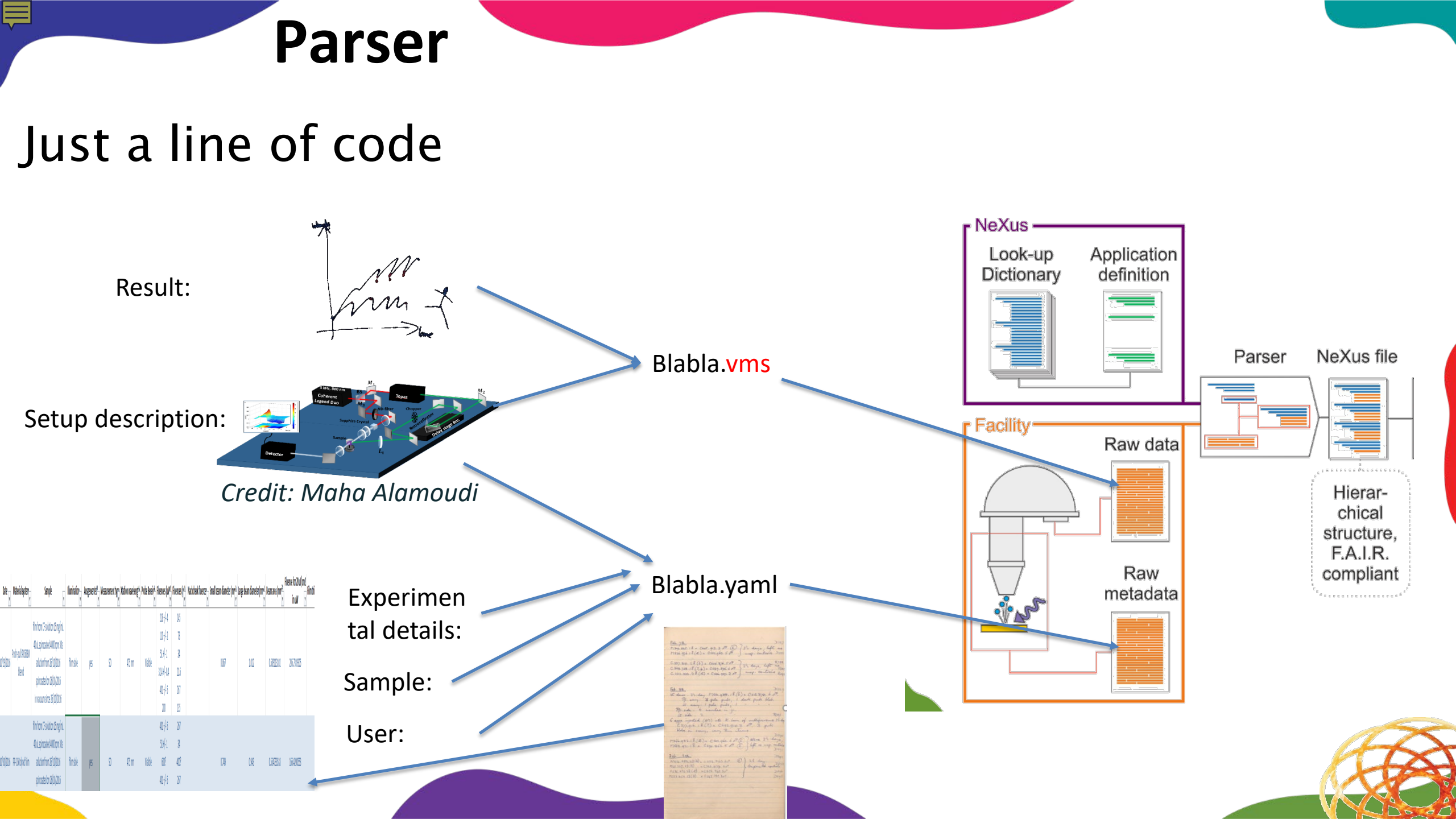
NeXus file

Hierarchical structure, F.A.I.R. compliant

Facility

Raw data

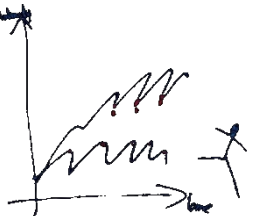
Raw metadata

[illegible]

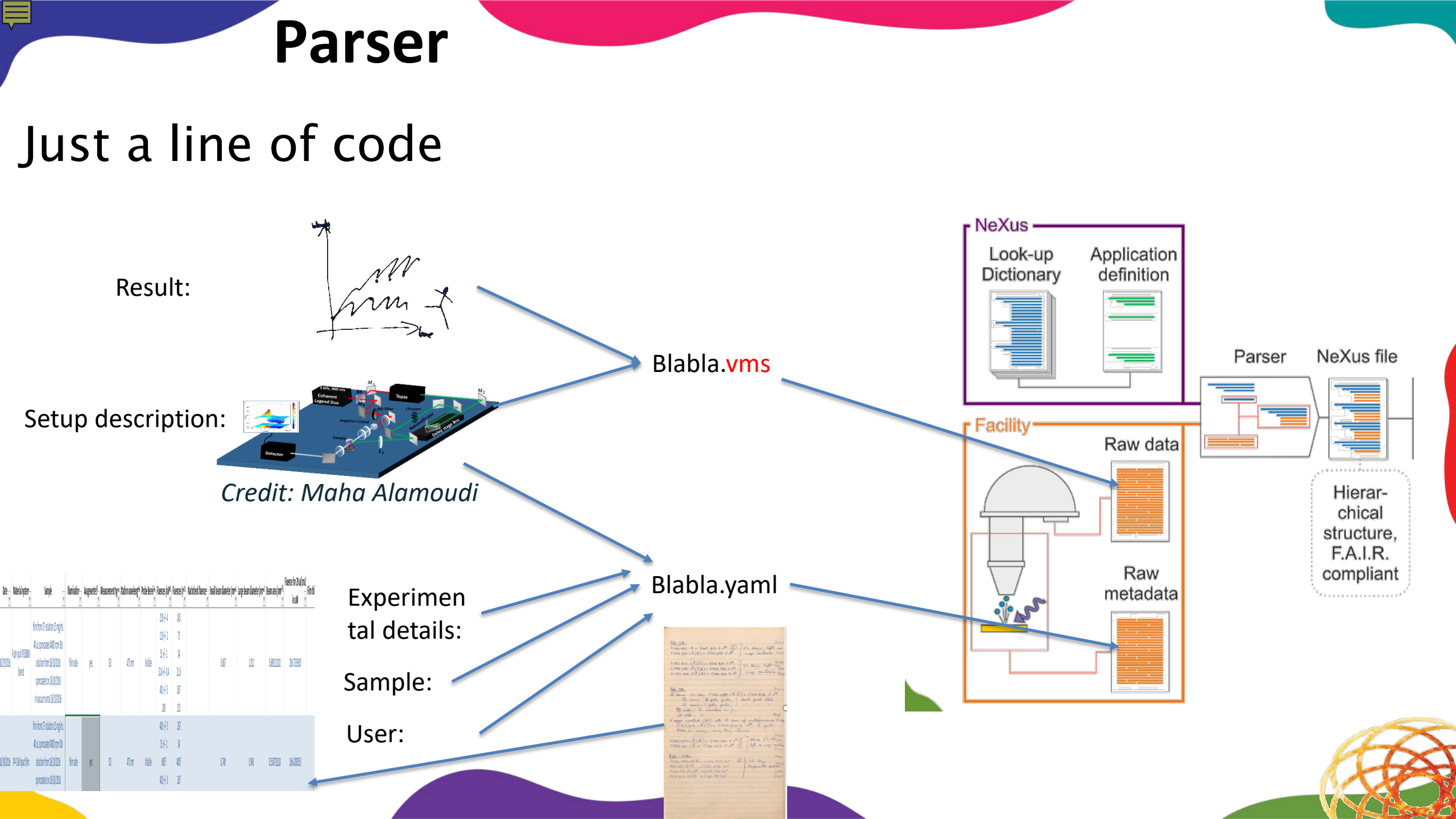
Parser

Just a line of code

Result:



Setup description:



Parser

Just a line of code

Result:

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Facility

Raw data

Raw metadata

[illegible]

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NeXus

Look-up Dictionary

Application definition

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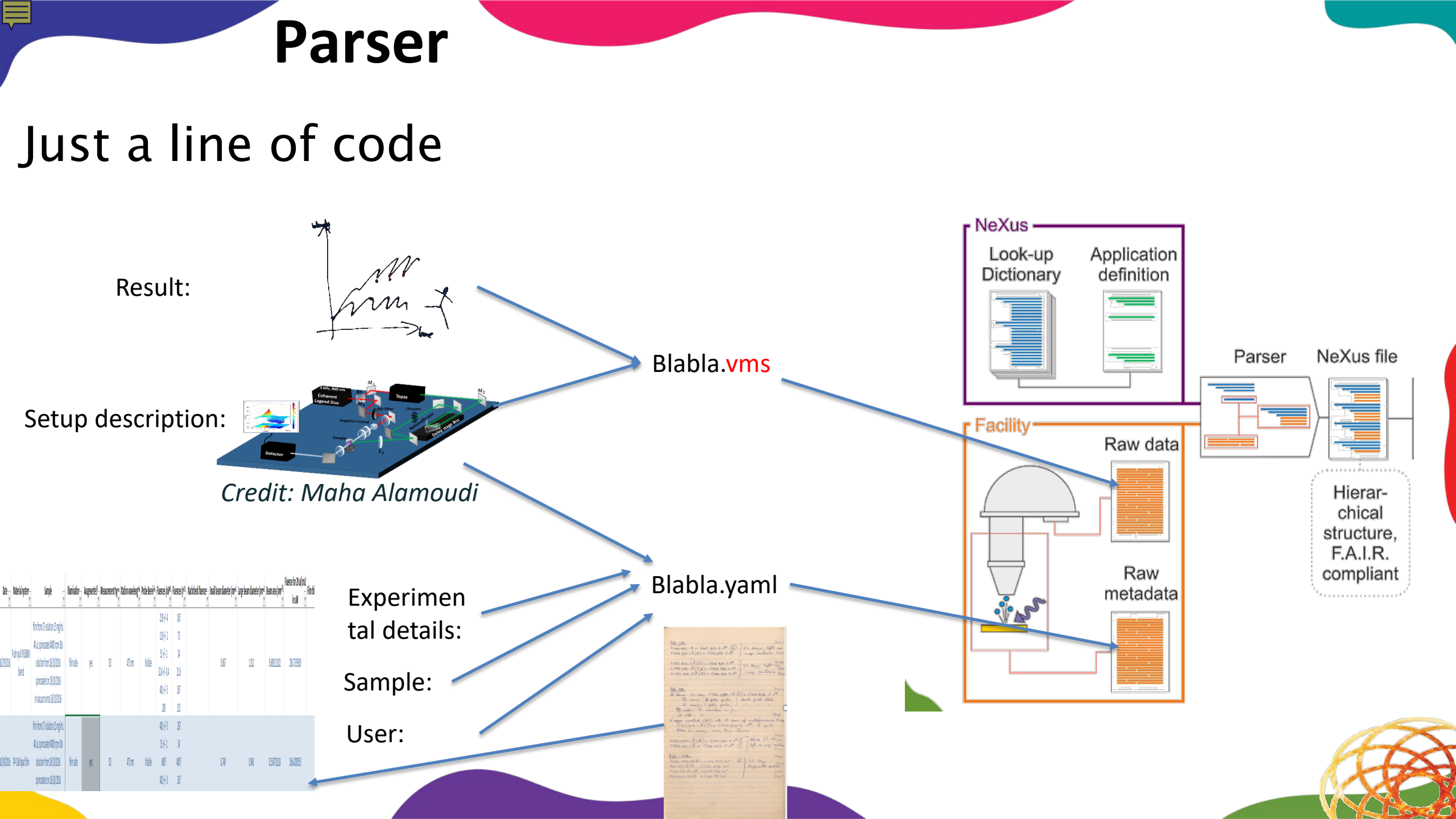
NeXus file

Hierarchical structure, F.A.I.R. compliant

Facility

Raw data

Raw metadata



GUI interface

FileDropMultiple

Drag and drop files(s) here to read the first 100 bytes.

! No files uploaded yet.

UPDATE ELN FILE

GENERATE NXS FILE



UPLOAD_TO_NOMAD_OASIS

And so on, and so on...

NOMAD = Just one solution among many (here ex. from material science)

- For ELN only : [cheminfo](#), [eLabTTW](#)
- Call for other examples (for international open access week?)

Other

We have a local NOMAD oasis:

10.75.90.88/nomad-oasis/gui

More resources:

- [NOMAD for Computational Material Science](#)
- [NOMAD Camels](#) -> get FAIR directly from your setup
- Much more on FAIRmat Youtube Channel

